

KRUNAL VAGHAMSHI

Data scientist

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[GITHUB Repo](#) 

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About Me

Passionate Data Science practitioner dedicated to the art of 'Vision-Full' engineering. I combine financial literacy with deep technical expertise to build systems that see, think, and predict. My focus is on architecting full-stack AI solutions—solving complex problems like cloud persistence and model optimization—to create intelligent, user-centric software that is as robust as it is innovative

PROJECTS

Niraakshan:

Multi-Target Guard Surveillance Ecosystem An advanced security suite replacing manual monitoring with AI-driven vigilance.

The Vision: Engineered a real-time tracking system to automate security protocols and detect negligence without human intervention.

The Tech: Python, MediaPipe Pose, Face Recognition, CustomTkinter, CLAHF (Night Mode).

Key Innovation: Developed independent multi-thread tracking for multiple guards simultaneously and a "Fugitive Mode" that triggers immediate audio alerts upon identifying specific targets.

Impact: Successfully implemented "Action Classification" to detect sleeping or inactivity and integrated auto-snapshot logging for accountability

CineMind AI: Semantic Movie Recommender

Tech: Python, Hugging Face Transformers, Google Sheets API, Streamlit.

Description: Built a recommendation engine that moves beyond keyword matching to understand user "mood" and semantic meaning.

Impact: Solved the "stateless container" data loss problem by architecting a custom connection to Google Sheets API for real-time, persistent database management without SQL costs.

NYC Smart Taxi Fare Predictor

Tech: XGBoost, PyDeck (3D Visualization).

Description: Built a spatial mapping tool to predict taxi fares based on location and time.

Impact: Achieved 97% model accuracy and implemented 3D Arc Layer visualizations to map pickup/drop-off density across New York City.

SleepAI Analytics: End-to-End Health Intelligence System

The Vision: Developed a comprehensive AI system to diagnose sleep disorders and predict sleep quality by analyzing lifestyle and cardiovascular metrics, pivoting wellness programs toward targeted intervention.

The Tech: Python, Random Forest, XGBoost, Streamlit, SMOTE, Power BI.

Key Innovation: Built a Dual-Model Predictive Core achieving 93.75% accuracy for disorder classification and 98% accuracy for continuous Sleep Quality prediction. Utilized proprietary features like the Health_Risk_Score.

Impact: Confirmed Stress as the dominant predictor of poor sleep. Deployed a Streamlit "Virtual Sleep Doctor" for real-time diagnosis and context-aware recommendations, supported by a Power BI dashboard for population risk analysis.

EDUCATION

Master of Computer

Applications (MCA) | Pursuing
(2024 - so on)

Swarnim Startup & Innovation
University

Data Science Certification
(2024 - so on)

Red & White Skills Education
Focus: Data Science, AI/ML,
and End-to-End Project
Development.

Bachelor of Commerce
(B.Com) (2021-2024)

SDJ International College

LANGUAGE

English

Hindi

Gujarati

SOFT SKILLS

Critical Thinking

Analytical Problem-Solving

Cross-Functional Collaboration

Adaptability

Detail-Oriented.

Technical Skills

Programming & Core Languages

Python (Advanced Modules, OOP), C/C++ (Foundational Knowledge), GitHub, Data Structures, Algorithms.

Data Science & Machine Learning

Pandas, NumPy, Scikit-learn, Data Pre-processing (Handling Missing Data, Outliers), Feature Engineering, Supervised Learning (Linear Regression, SVM, KNN), Model Evaluation (MSE, R^2 Score, Confusion Matrix, F1 Score), Regularization (Ridge, Lasso), Ensemble Methods.

Deep Learning

Deep Learning (ANN, CNN, RNN/LSTM), TensorFlow/Keras (Implied from topics), Transformers, Backpropagation, Weight Initialization, Batch Normalization, Optimization (SGD, Adam), Transfer Learning.

Data Analysis & Databases

SQL (CRUD, Joins, GroupBy), Microsoft Power BI (DAX, Data Modeling, Visualization), Matplotlib, Seaborn, Exploratory Data Analysis (EDA).

Quantitative Foundations

Statistics (Inferential & Descriptive), Hypothesis Testing (T-Test, Z-Test, ANOVA), Probability, Linear Algebra (PCA, SVD, Matrix Operations).

Auxiliary Tools & Concepts

Computer Vision (OpenCV), Web Scraping (Beautiful Soup), Streamlit (Full-Stack), Google Sheets API, Regular Expressions, File Handling, Exception Handling, Command Line Arguments.